ZHANG Bin, Ph.D. Candidate

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Education

Sep. 2021 – Now	**Ph.D. Student, The Hong Kong Polytechnic University** in *Mechanical Engineering*.
Mar. 2024 – Jul. 2024	**Visiting Ph.D. Student, KTH Royal Institute of Technology**.
Sep. 2017 – Jul. 2020	**M.Sc., China Academy of Space Technology** in *Control Science and Engineering*.
Sep. 2013 – Jul. 2017	**B.Eng., Beihang University** in *Detection, Guidance, and Control Technology*.

Employment History

May 2021 – Aug. 2021	**Research Assistant** at ROMI Lab, The Hong Kong Polytechnic University.

Research Interests

- Multi-agent formation control.
- Multi-robot manipulation.
- Control theory and applications.

Research Publications

* Equal contribution

Journal Articles

1. X. Shao*, **B. Zhang***, H. Zhi, *et al.*, "Inptc: Integrated planning and tube-following control for prescribed-time collision-free navigation of wheeled mobile robots," *IEEE/ASME Transactions on Mechatronics*, 2024, (Early Access). DOI: 10.1109/TMECH.2024.3486727. URL: https://vimeo.com/895801720.
2. Y. Tian, **B. Zhang**, X. Shao, and D. Navarro-Alarcon, "Encircling general 2-d boundaries by mobile robots with collision avoidance: A vector field guided approach," *IEEE/ASME Transactions on Mechatronics*, 2024, (Under Review). DOI: arXiv:2501.02242.
3. **B. Zhang**, H. Zhi, J. G. Romero, and D. Navarro-Alarcon, "Leader-follower formation control of perturbed nonholonomic agents along parametric curves with directed communication," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8603–8610, Aug. 2024. DOI: 10.1109/LRA.2024.3445657. URL: https://vimeo.com/872927697.
4. H. Zhi, **B. Zhang**, J. Qi, *et al.*, "Non-prehensile object transport by nonholonomic robots connected by linear deformable elements," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8651–8658, Aug. 2024. DOI: 10.1109/LRA.2024.3440096.
5. H. Y. Lee, P. Zhou, **B. Zhang**, *et al.*, "A distributed dynamic framework to allocate collaborative tasks based on capability matching in heterogeneous multi-robot systems," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 16, no. 1, pp. 251–265, Feb. 2023. DOI: 10.1109/TCDS.2023.3264034.

6. W. Ma, **B. Zhang**, L. Han, S. Huo, H. Wang, and D. Navarro-Alarcon, "Action planning for packing long linear elastic objects into compact boxes with bimanual robotic manipulation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 3, pp. 1718–1729, Jun. 2023. DOI: 10.1109/TMECH.2022.3223990.

7. M. Muddassir, G. Limbert, **B. Zhang**, A. Duan, J.-J. Tan, and D. Navarro-Alarcon, "Model predictive thermal dose control of a robotic laser system to automate skin photorejuvenation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 2, pp. 737–747, Apr. 2023. DOI: 10.1109/TMECH.2022.3218806.

8. **B. Zhang**, H. Zhi, J. G. Romero, *et al.*, "Fourier-based multi-agent formation control to track evolving closed boundaries," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 11, pp. 4549–4559, Nov. 2023. DOI: 10.1109/TCSI.2023.3300772. URL: https://vimeo.com/809466573.

9. S. Huo, **B. Zhang**, M. Muddassir, D. T. W. Chik, and D. Navarro-Alarcon, "A sensor-based robotic line scan system with adaptive roi for inspection of defects over convex free-form specular surfaces," *IEEE Sensors Journal*, vol. 22, no. 3, pp. 2782–2792, Feb. 2022. DOI: 10.1109/JSEN.2021.3132428.

10. J. Zhou, J. Hu, Y. Bai, and **B. Zhang**, "Optimal impulsive time-fixed transfers around the libration points of the restricted three-body problem," *Astrophysics and Space Science*, vol. 365, pp. 1–18, May 2020. DOI: 10.1007/s10509-020-03793-7.

11. J. Zhou, J. Hu, and **B. Zhang**, "Analytical solutions for relative motion in the cicular restricted three-body problem," *Journal of Astronautics*, vol. 41, no. 2, Feb. 2020, (in Chinese). DOI: 10.3873/j.issn.1000-1328.2020.02.004.

12. J. Zhou, J. Hu, and **B. Zhang**, "Approximate analytical solutions of motion near the collinear libration-points in restricted three-body problem," *Journal of Deep Space Exploration*, vol. 7, no. 1, Feb. 2020, (in Chinese). DOI: 10.15982/j.issn.2095-7777.2020.20190408001.

13. **B. Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbit," *Acta Aeronautica et Astronautica Sinica*, vol. 40, no. 11, Nov. 2019, (in Chinese). DOI: 10.7527/S1000-6893.2019.23306.

Conference Proceedings

1. A. Duan, **B. Zhang**, L. Hu, L. Feng, and D. Navarro-Alarcon, "Balancing control of wheeled-legged hexapedal robots for traversing rough terrains," in *2022 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Dec. 2022, pp. 1905–1910. DOI: 10.1109/ROBIO55434.2022.10011895.

2. **B. Zhang**, C. Li, and J. Zhou, "Station-keeping of halo orbits using convex optimization-based receding horizon control," in *2021 China Automation Congress (CAC)*, Oct. 2021, pp. 7160–7165. DOI: 10.1109/CAC53003.2021.9728345.

3. **B. Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbits about sun-earth l1 point," in *2019 Chinese Automation Congress (CAC)*, Nov. 2019, pp. 418–423. DOI: 10.1109/CAC48633.2019.8997351.

Skills

Languages	English, Chinese (*mother tongue*).
Coding	Python, C/C++, Matlab, HTML, LaTeX.
Others	ROS, Linux, Webots, Arduino, SolidWorks.

Miscellaneous Experience

Awards and Achievements

2017 **Outstanding Graduate of Beihang University.**

2015 **National Inspirational Scholarship.**

2014 **National Inspirational Scholarship.**

Academic Services

2021 **Reviewer**. ICRA 2022.

2023 **Reviewer**. ICRA 2024.

Reviewer. IEEE Control Systems Letters.

2024 **Reviewer**. Journal of Field Robotics.

Reviewer. IEEE Robotics and Automation Magazine.

Reviewer. IEEE Transactions on Control of Network Systems.

Reviewer. CASE 2024.

Reviewer. ICRA 2025.

Reviewer. Aerospace Science and Technology.

Teaching Experience

2021 **Teaching Assistant**, Perceptual Robotics.

2022 **Teaching Assistant**, Linear Systems and Control.

Teaching Assistant, Advanced Control Technology.

2023 **Teaching Assistant**, Linear Systems and Control.